OZ Lab

est. August 2024

#3 / Cost Functions and Gradient Descent

Understanding the Cost Function

@oz.lab__

Simplifying to One Parameter

model:

$$f_{w,b}(x) = wx + b$$

parameters:

w, b



cost function:

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

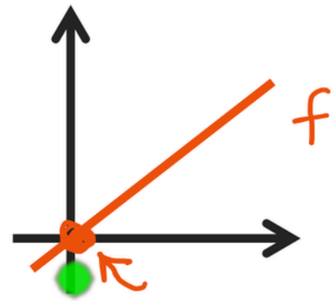
$$\underset{w,b}{\text{minimize}}\, J(w,b)$$

simplified

$$f_w(x) = wx$$

$b = \emptyset$

w

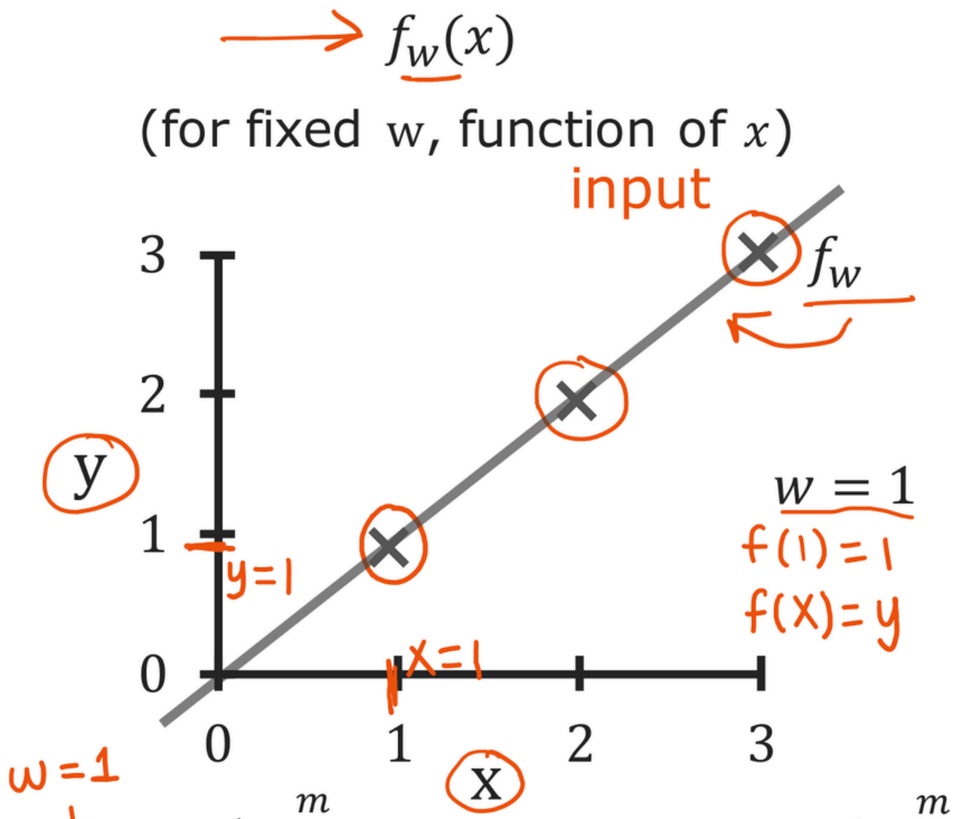


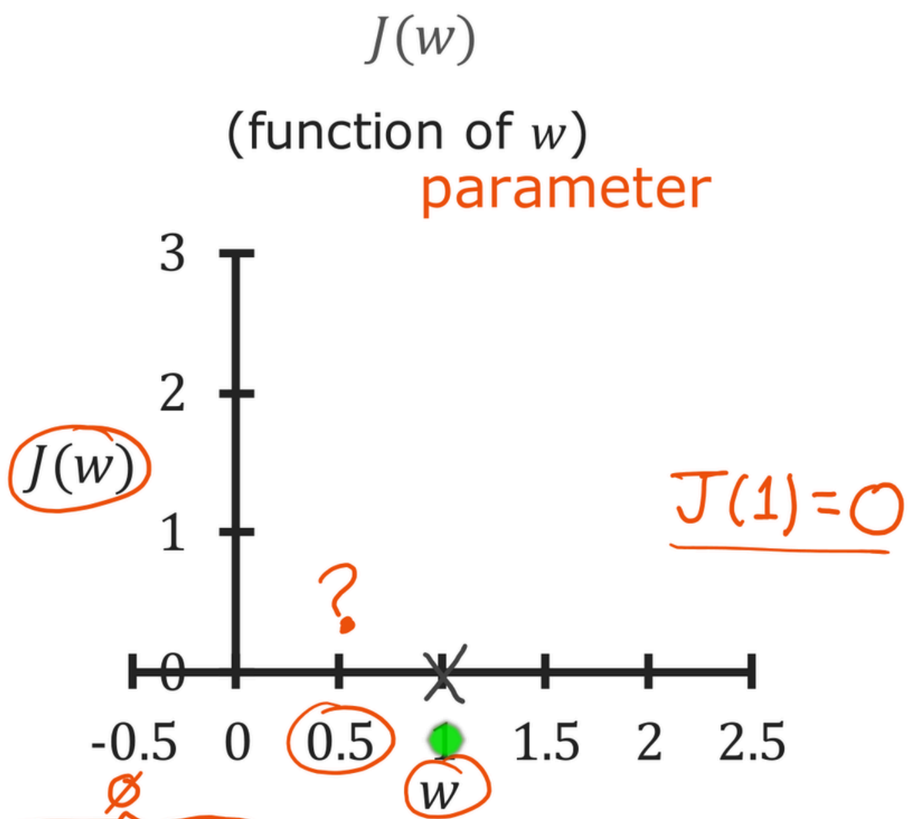
$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2$$

$wx^{(i)}$

$$\underset{w}{\text{minimize}}\, J(w)$$

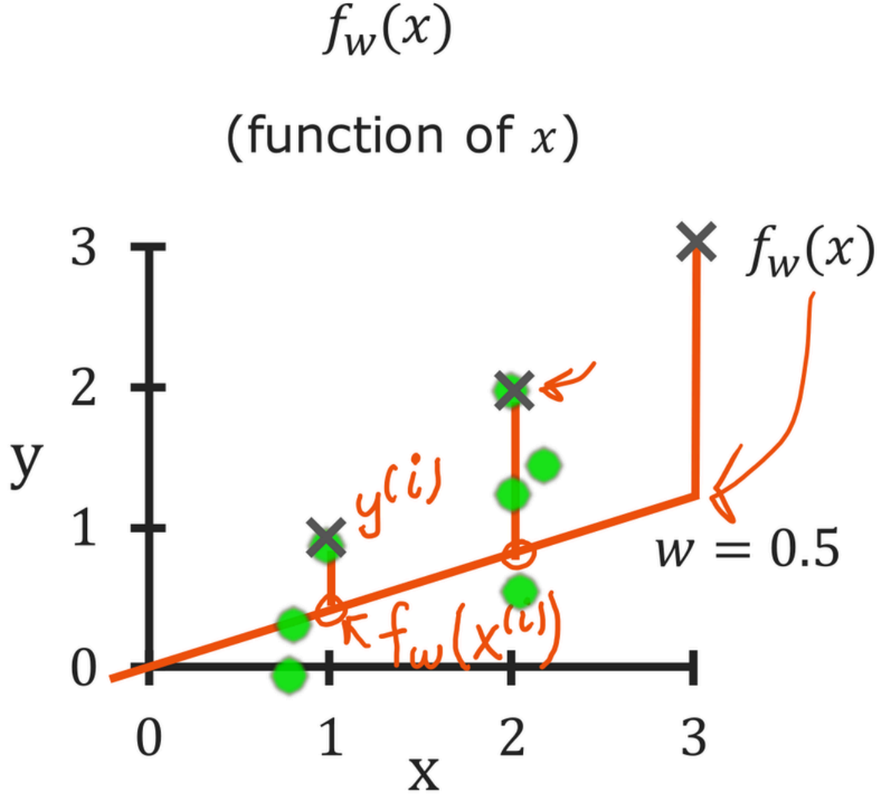
The Cost of a Perfect Fit

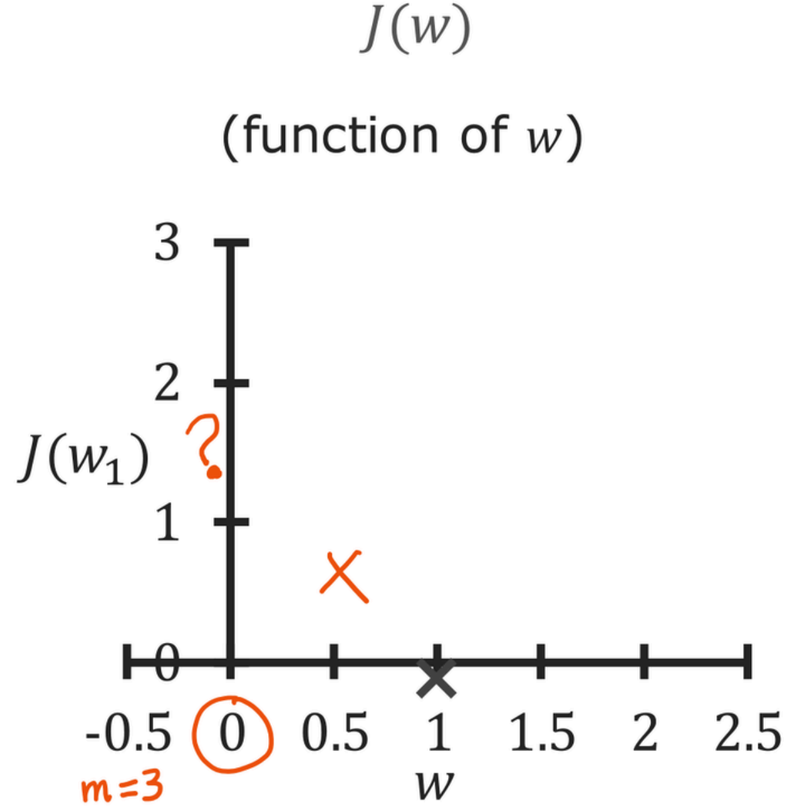




$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m}\sum_{i=1}^{m}(wx^{(i)} - y^{(i)})^2 = \frac{1}{2m}(0^2 + 0^2 + 0^2) = 0$$

The Cost of a Smaller Slope

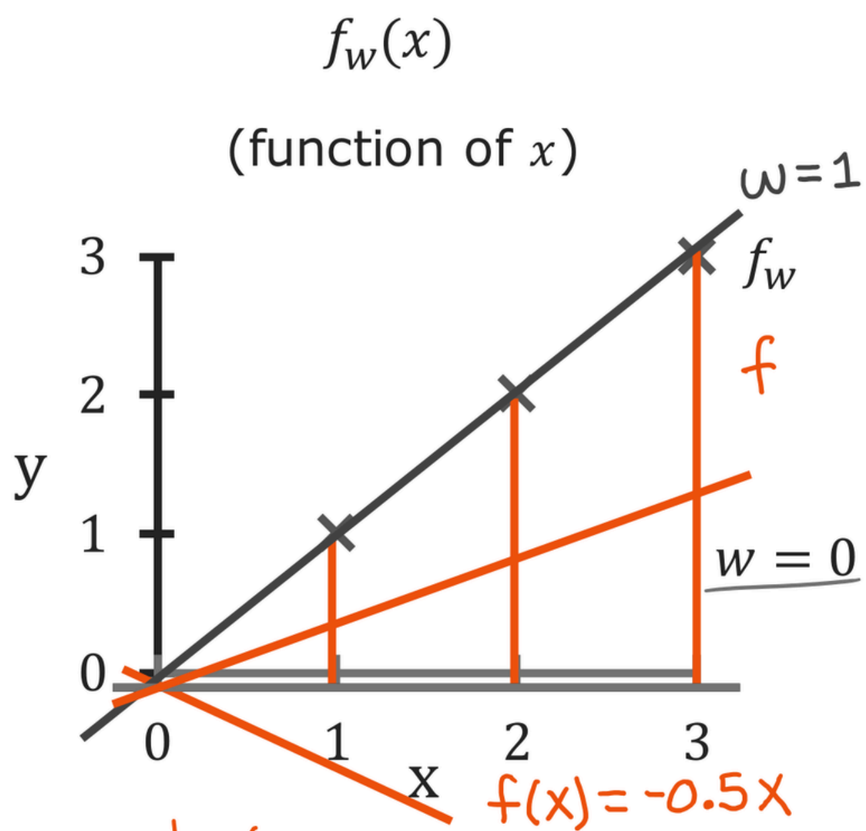


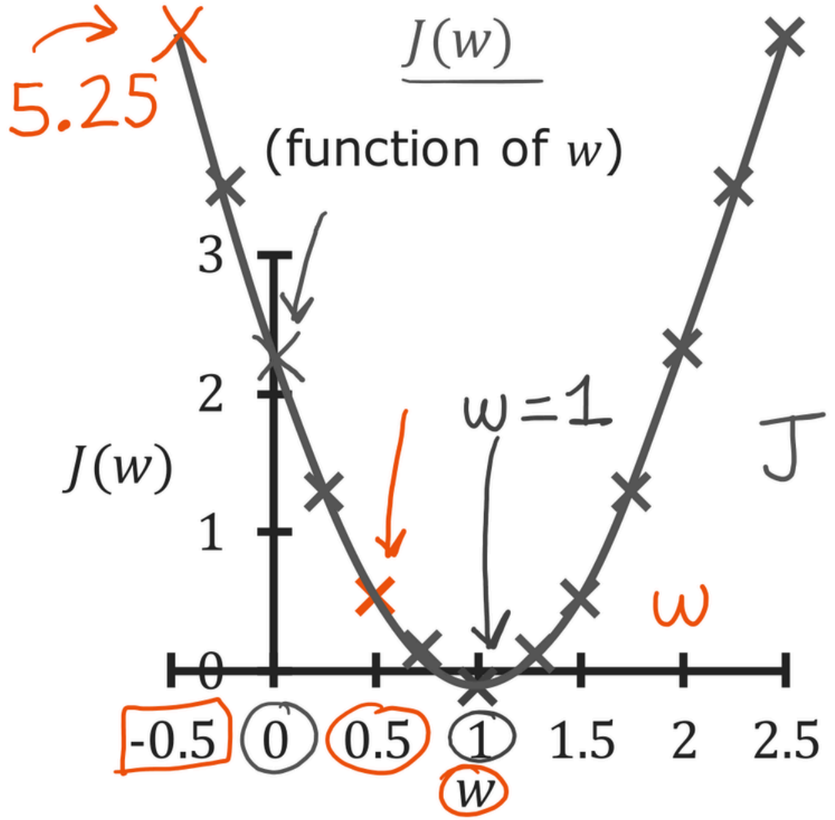


$$J(0.5) = \frac{1}{2m}\left[(0.5-1)^2+(1-2)^2+(1.5-3)^2\right] = \frac{1}{2\times 3}[3.5] = \frac{3.5}{6} \approx 0.58$$

m=3

Comparing Costs across Slopes





$$J(0) = \frac{1}{2m}\left(1^2+2^2+3^2\right) = \frac{1}{6}[14] \approx 2.3$$

how to choose w?

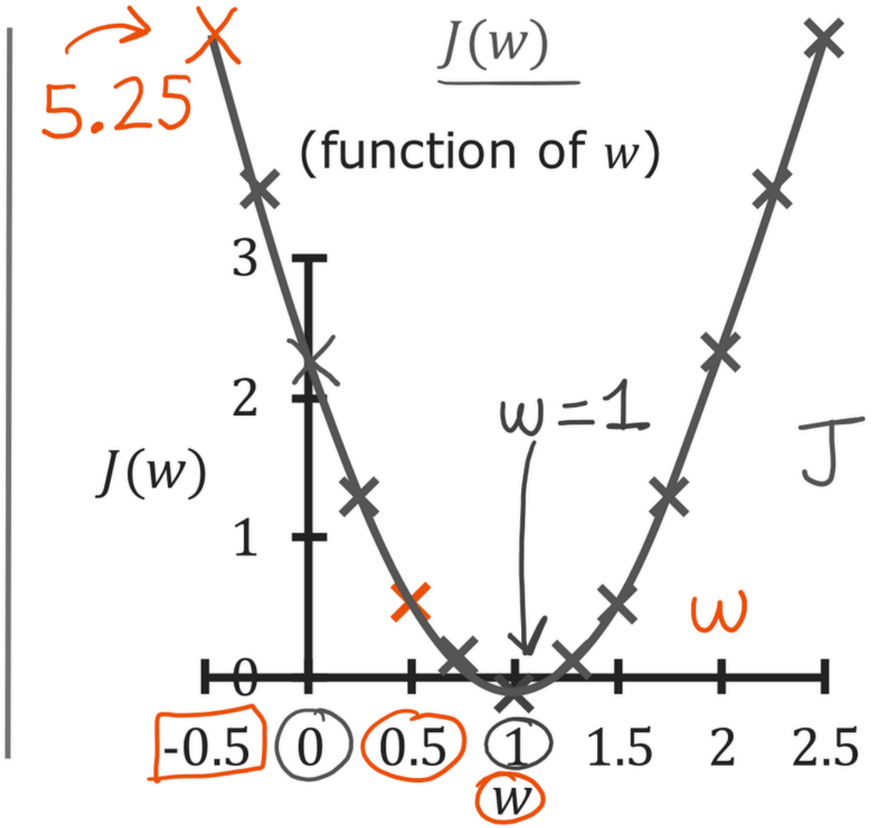
Finding the Lowest Cost

goal of linear regression:

$$\underset{w}{\text{minimize}}\, J(w)$$

general case:

$$\underset{w,b}{\text{minimize}}\, J(w,b)$$



choose w to minimize J(w)

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Visualizing the Cost Function

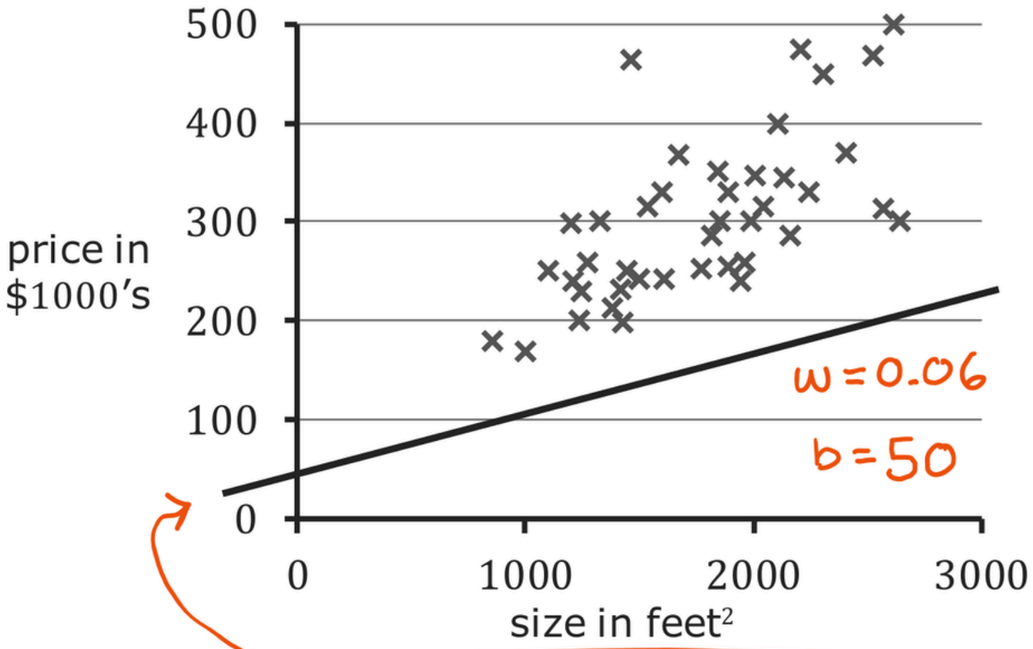
The Model, Cost, and Objective

Model	$f_{w,b}(x) = wx + b$
Parameters	before: ~~$b=0$~~ w, b
Cost Function	$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$
Objective	$\underset{w,b}{\text{minimize}}\, J(w,b)$

One Line, One Point on the Cost Surface

$f_{w,b}$

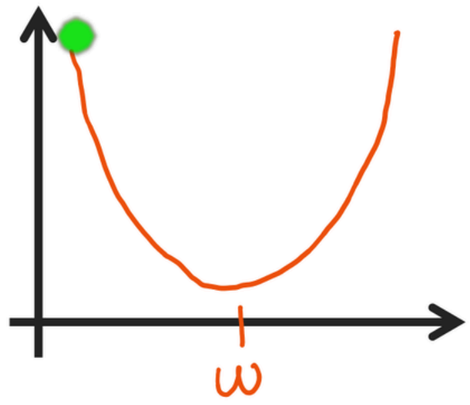
(function of x)



$f_{w,b}(x) = 0.06x + 50$

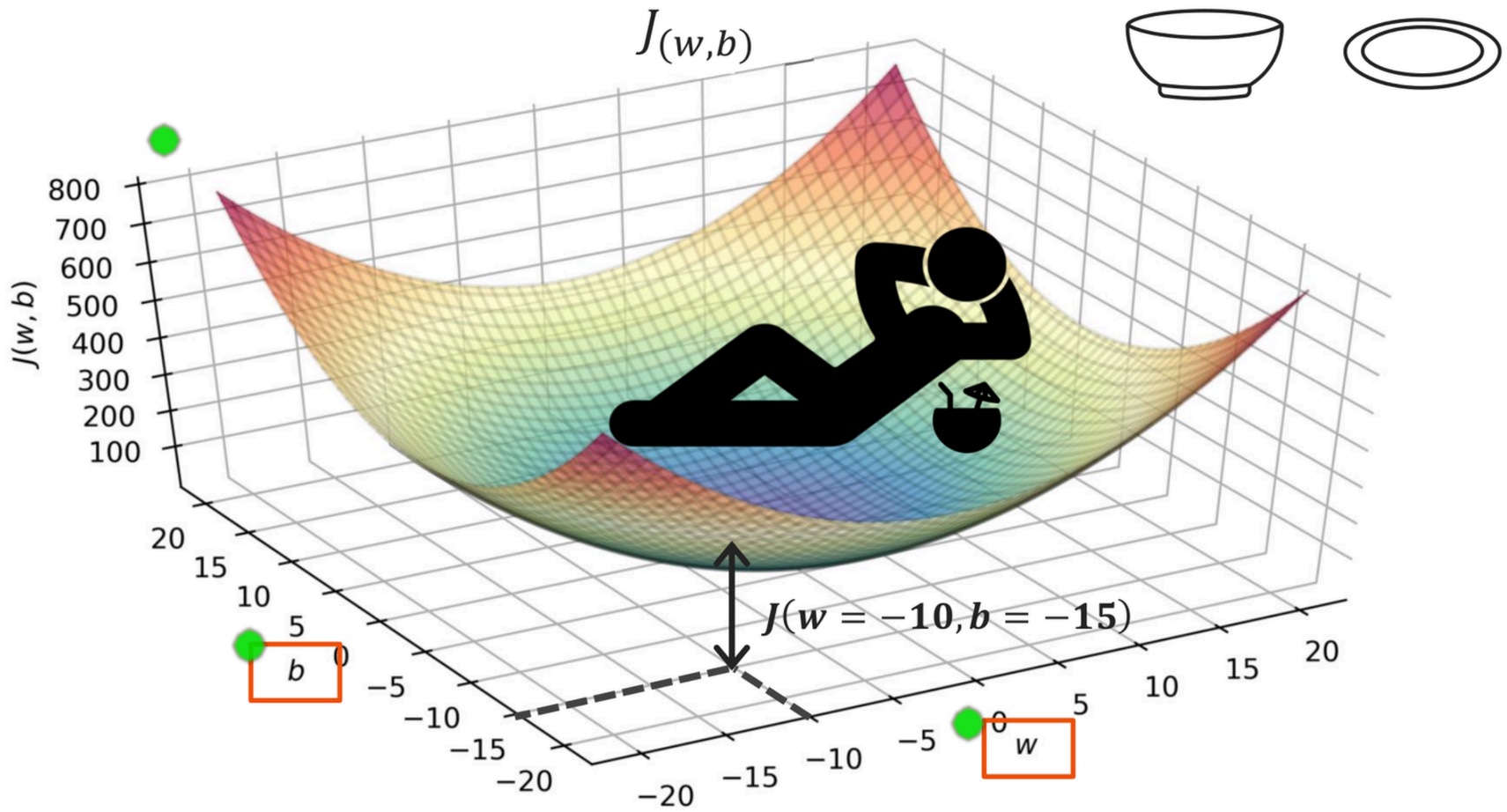
J

(function of w, b)

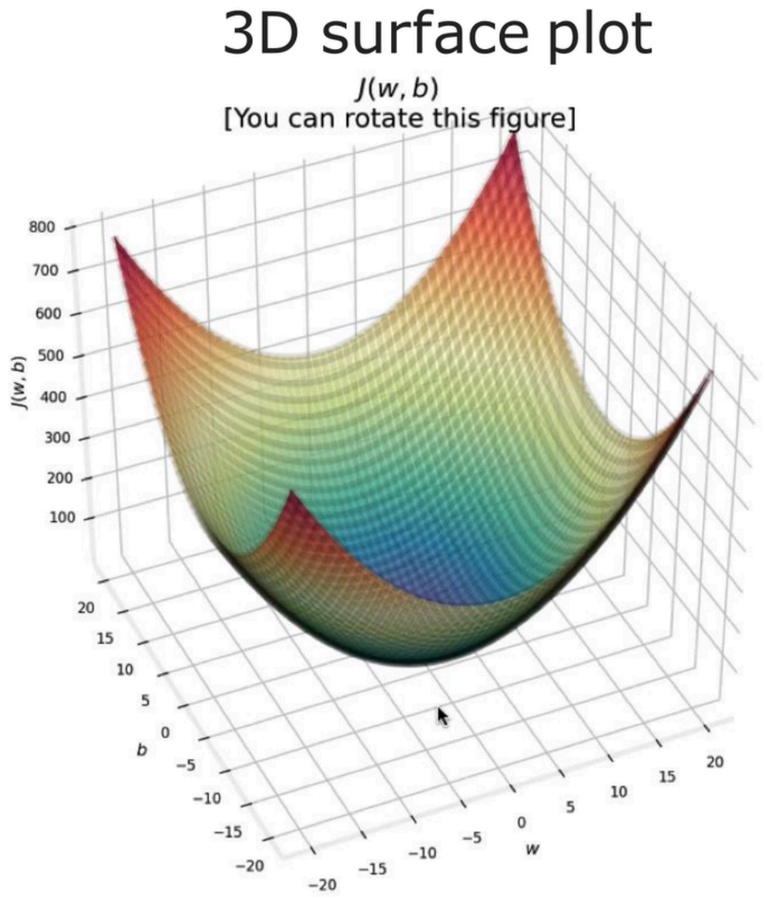


w , b

Cost as a Surface

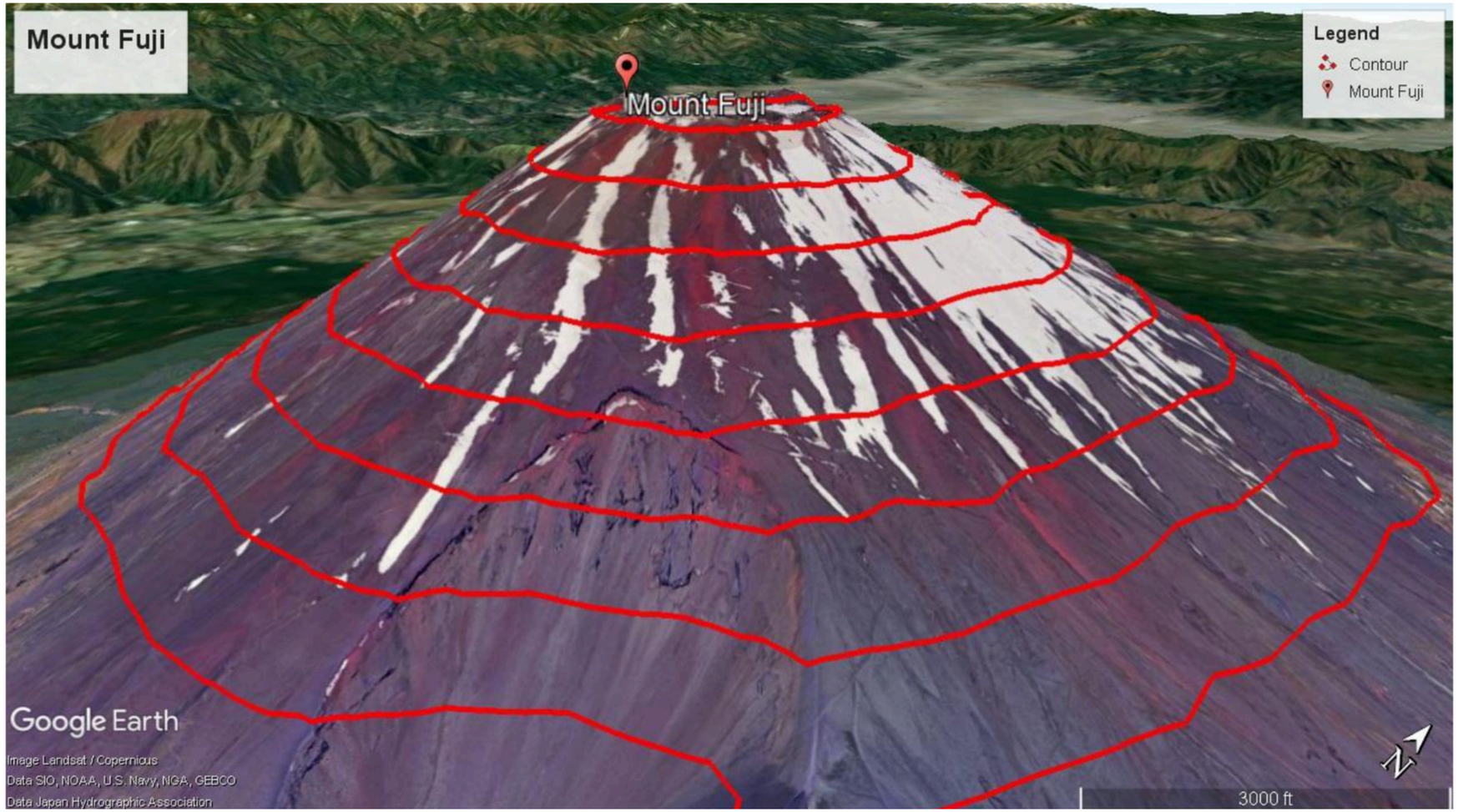


From a Surface to a Contour Plot

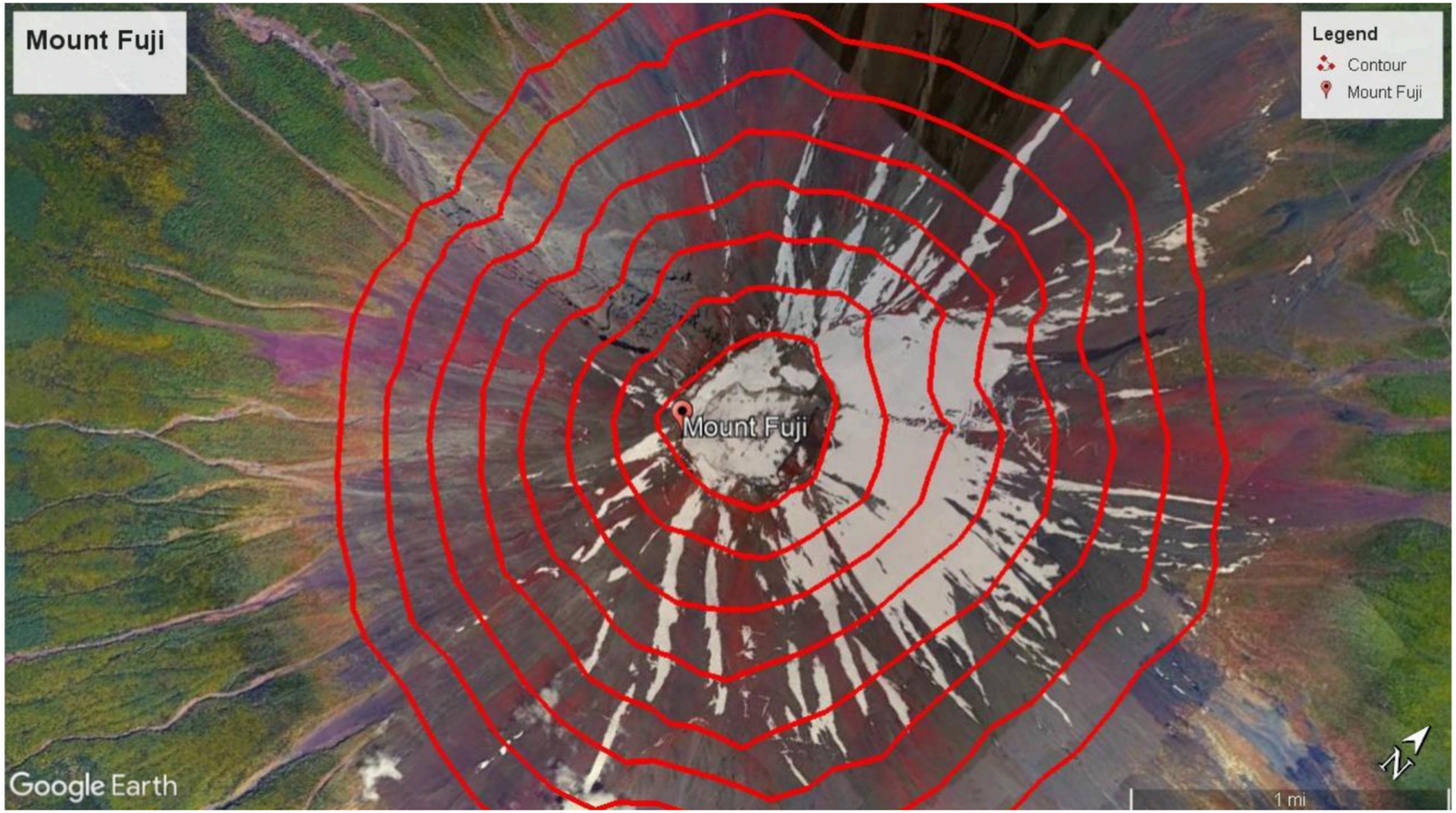


Alternative contour plot

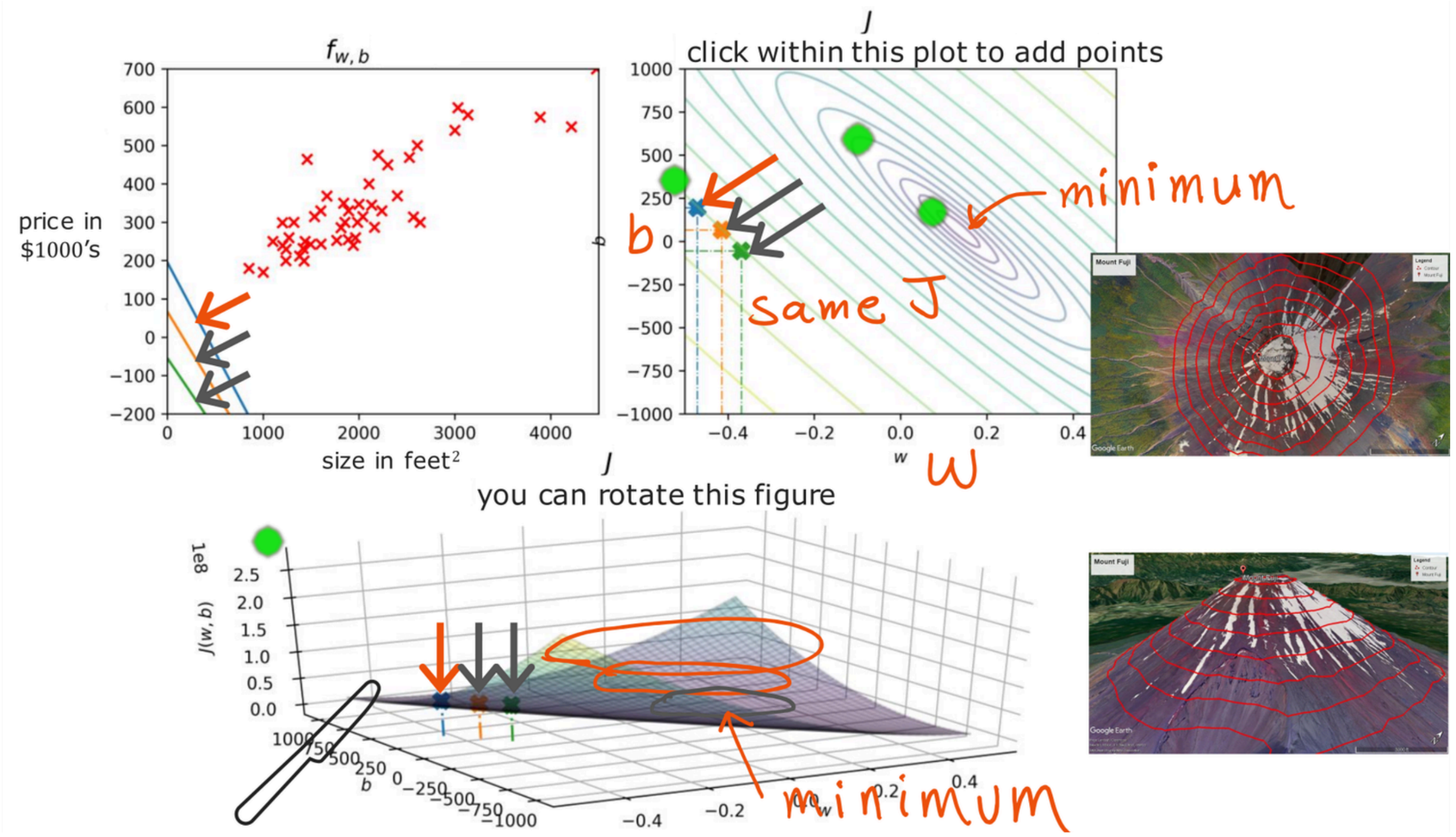
Contours Connect Equal Heights



Reading Contours from Above



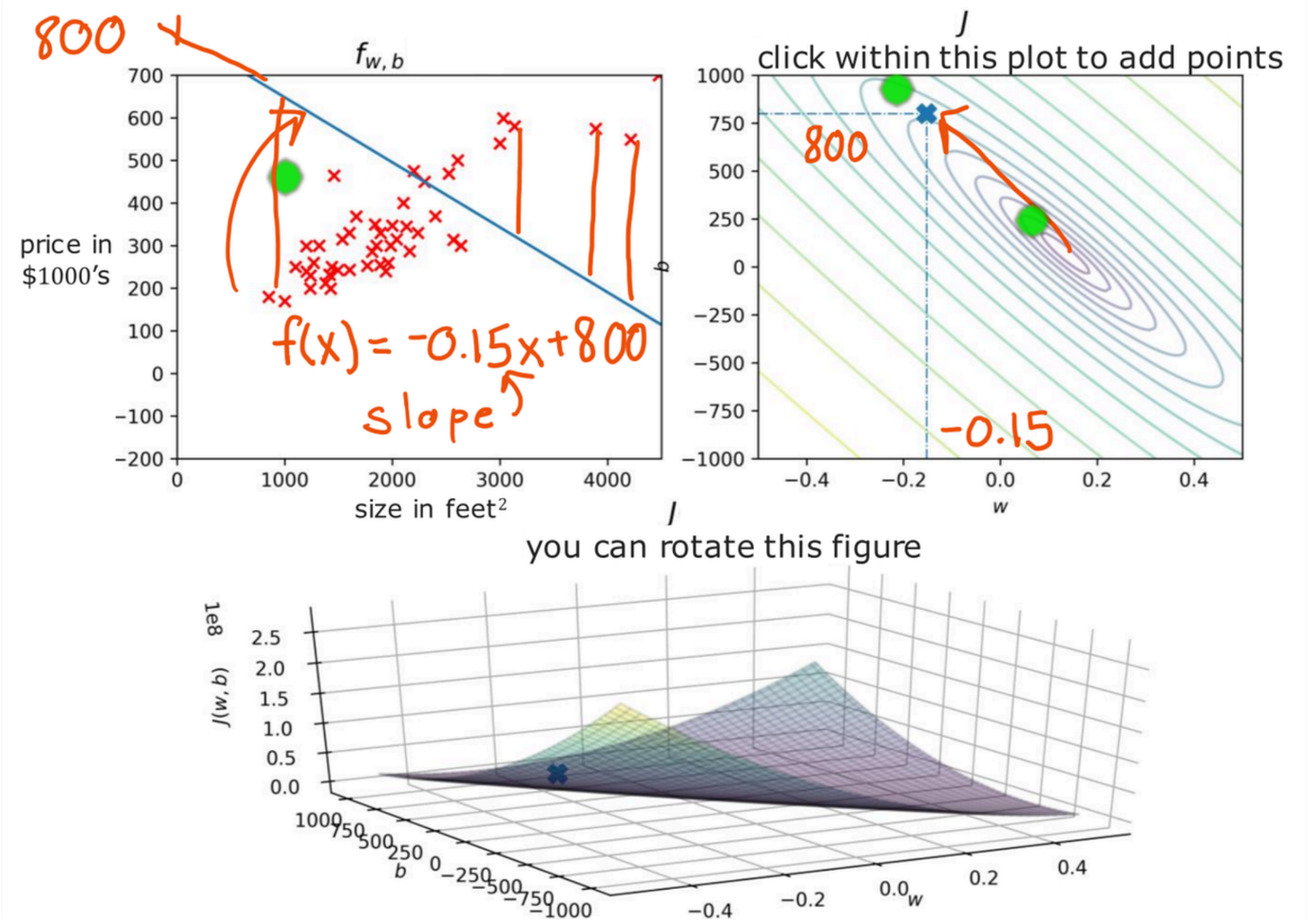
Connecting Lines, Surfaces, and Contours



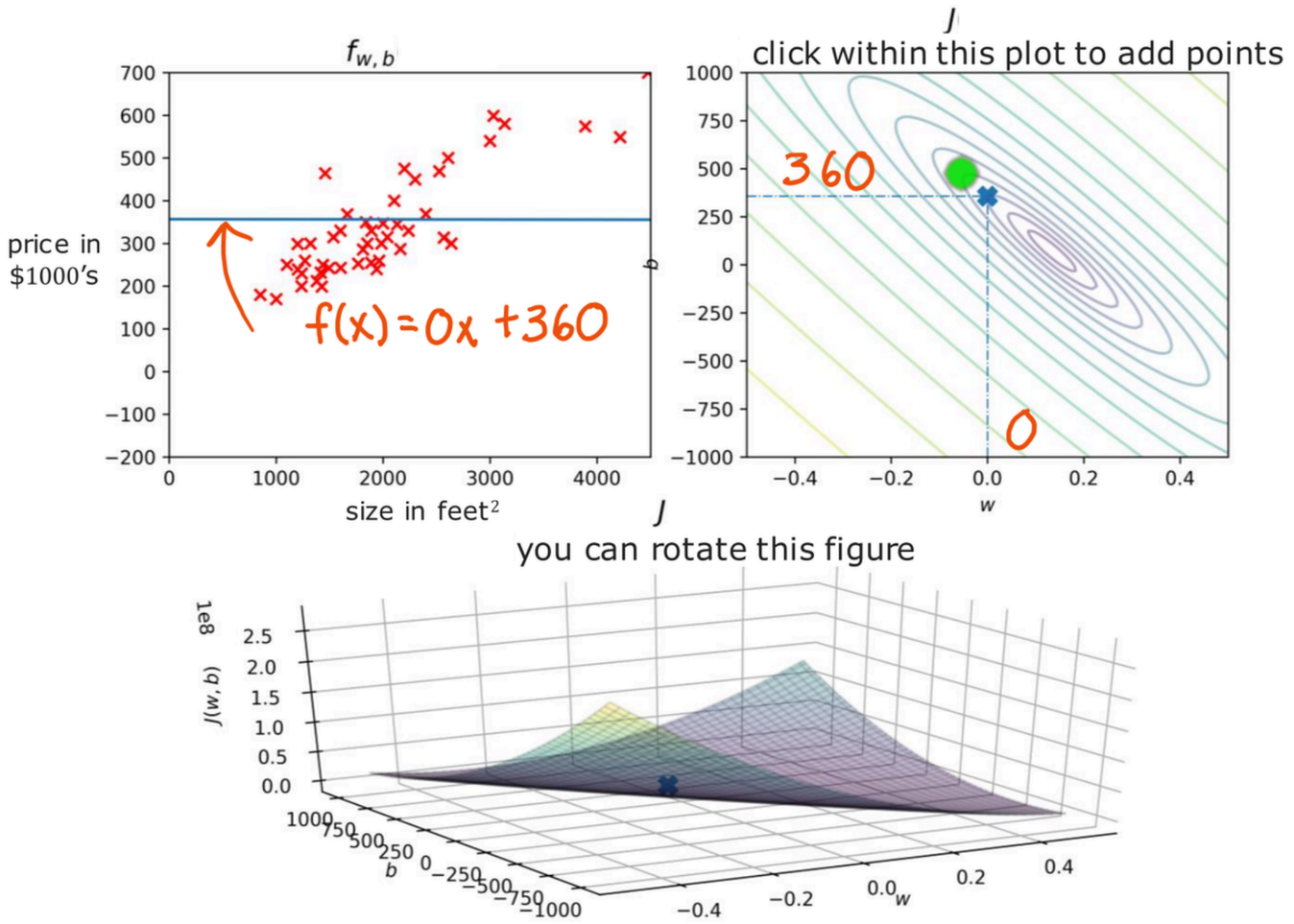
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Comparing Lines and Their Costs

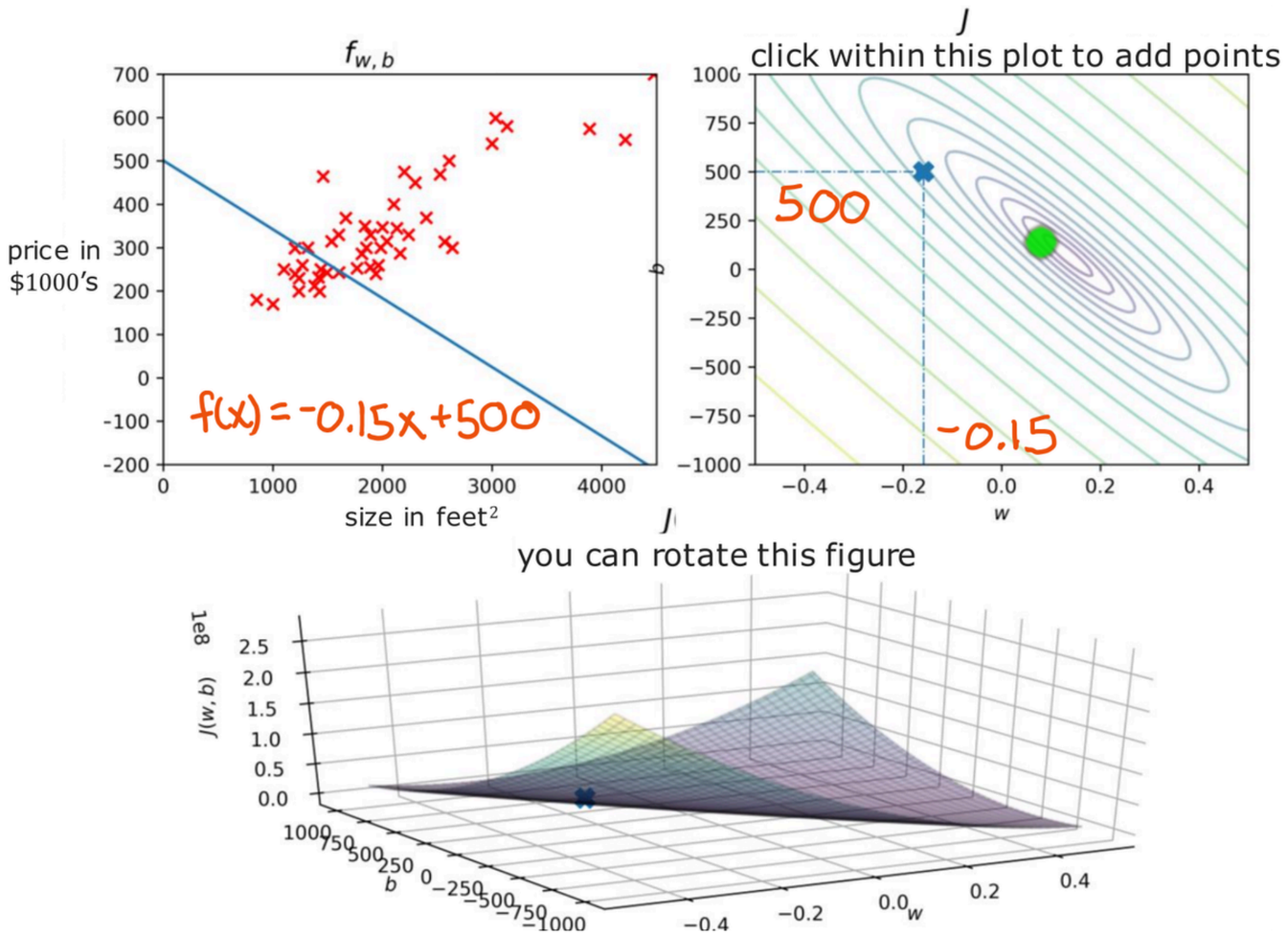
A Line with the Wrong Slope



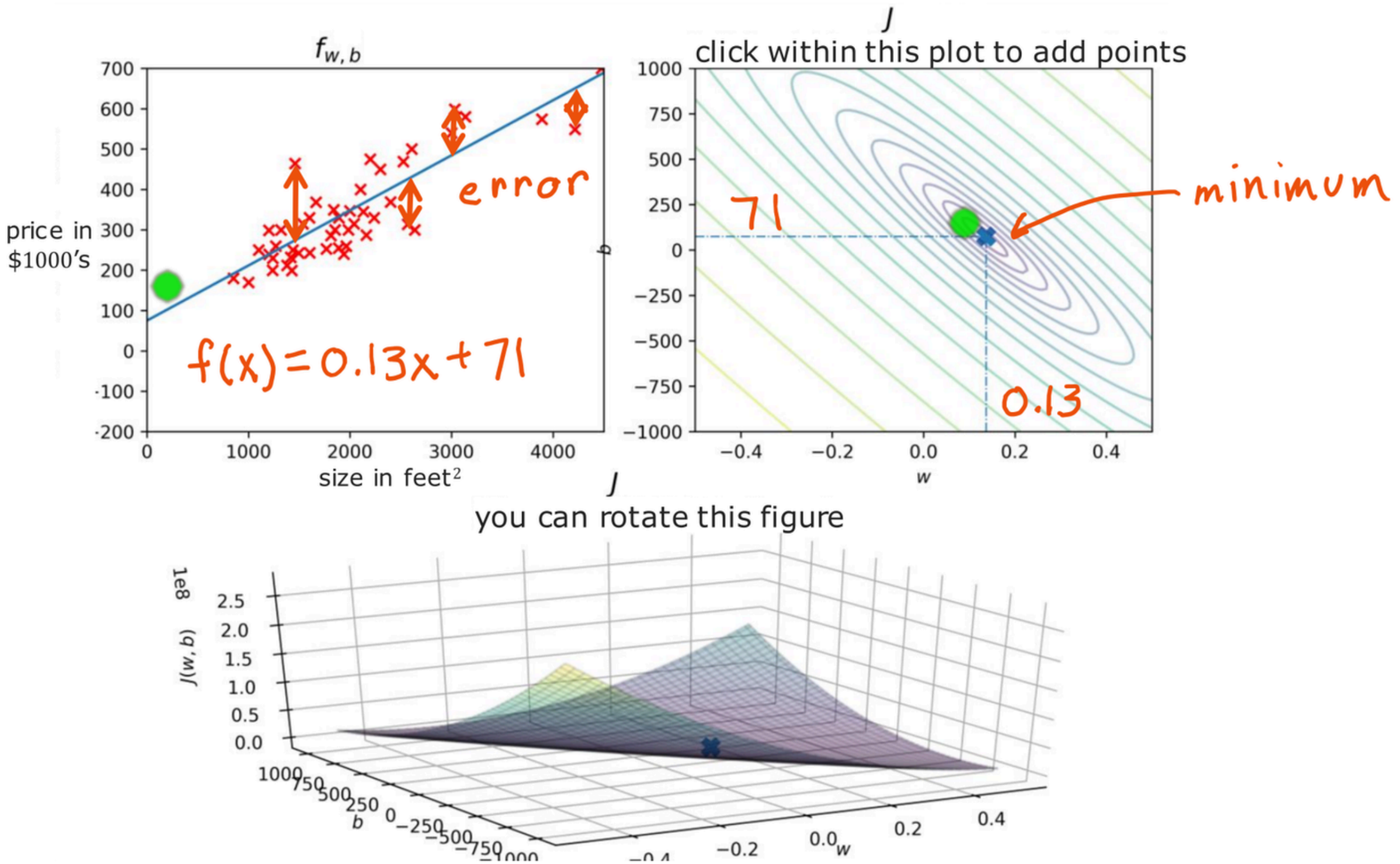
A Constant Price Prediction



Changing the Intercept



A Line That Fits the Trend



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Gradient Descent

Repeated Updates Reduce Cost

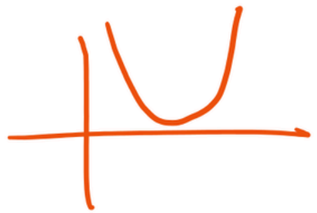
Have some function $J(w, b)$ for linear regression or any function

Want $\min_{w,b} J(w, b)$ $\min_{w_1, \ldots, w_n, b} J(w_1, w_2, \ldots, w_n, b)$

Outline:

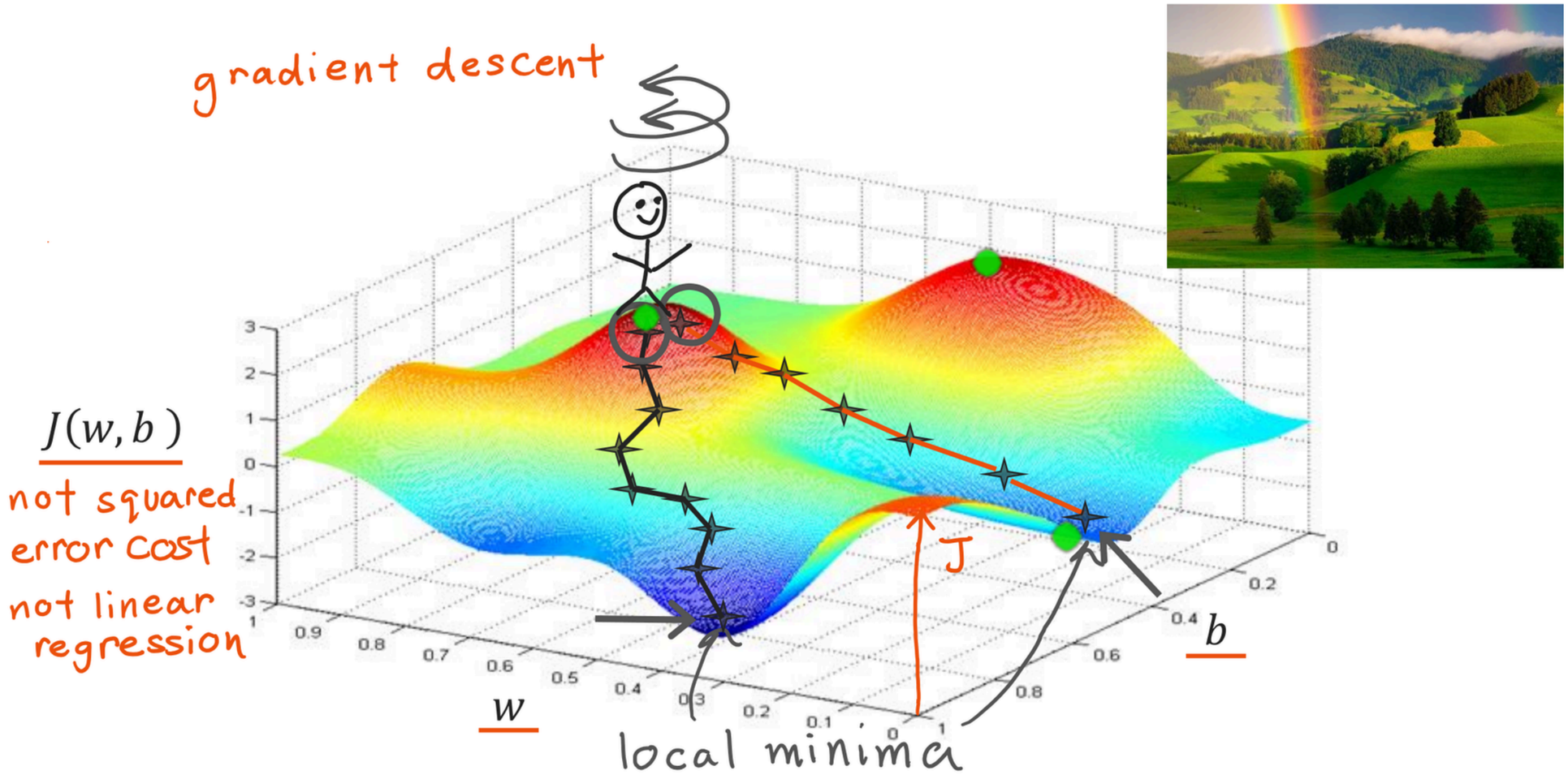
- Start with some w, b (set $w=0, b=0$)
- Keep changing w, b to reduce $J(w, b)$
- Until we settle at or near a minimum

J not always



may have >1 minimum

Different Starts, Different Minima



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Implementing Gradient Descent

Updating Parameters Simultaneously

Repeat until convergence

$$w = w - \alpha \frac{d}{dw} J(w,b)$$

$$b = b - \alpha \frac{d}{db} J(w,b)$$

Learning rate
Derivative

Simultaneously update w and b

Assignment

$a = c$

$a = a + 1$

Code

Truth assertion

$a = c$

$a = a + 1$ ✗

Math

a==c

Correct: Simultaneous update

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Incorrect

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$w = tmp_w$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$b = tmp_b$$

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Understanding Gradient Descent

Learning Rate and Derivative

repeat until convergence {

learning rate

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

derivative

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

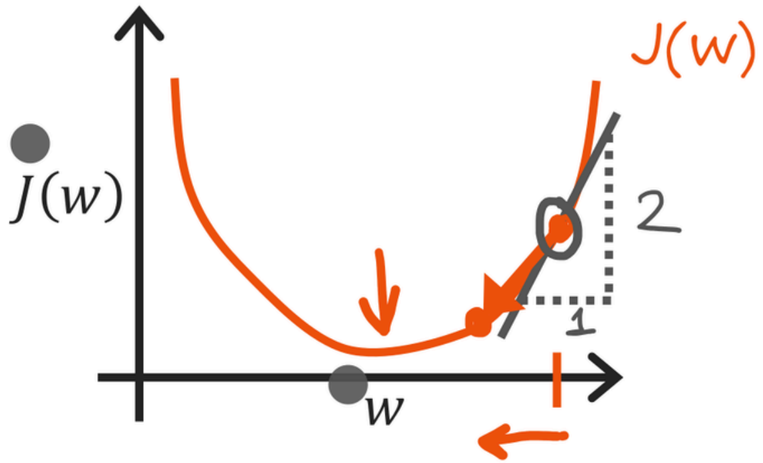
}

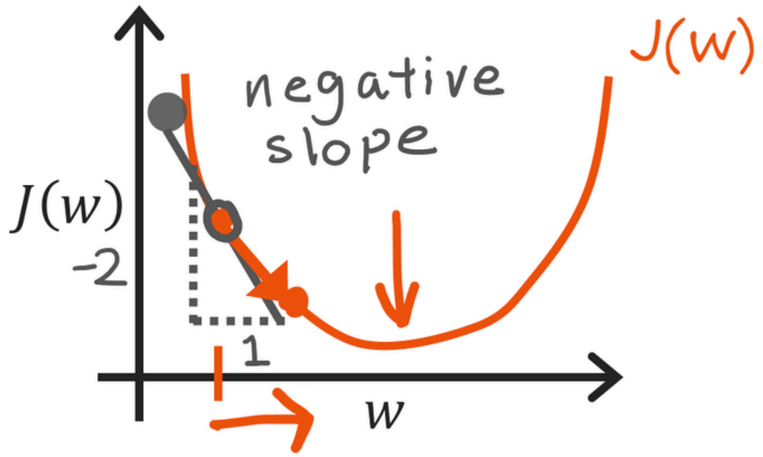
$$J(w)$$

$$w = w - \alpha \frac{\partial}{\partial w} J(w)$$

$$\min_{w} J(w)$$

Moving against the Slope





$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{>0}$$

$$w = w - \alpha \cdot (positive\ number)$$

$$\underbrace{\frac{d}{dw} J(w)}_{<0}$$

$$w = w - \alpha \cdot (negative\ number)$$

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The Learning Rate

Small and Large Learning Rates

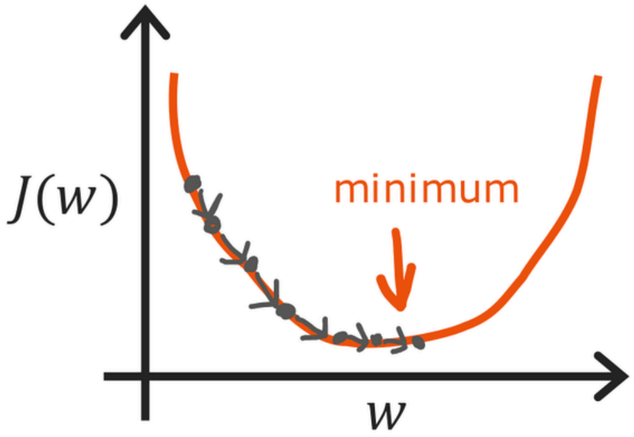
$$w = w - \alpha \frac{d}{dw} J(w)$$

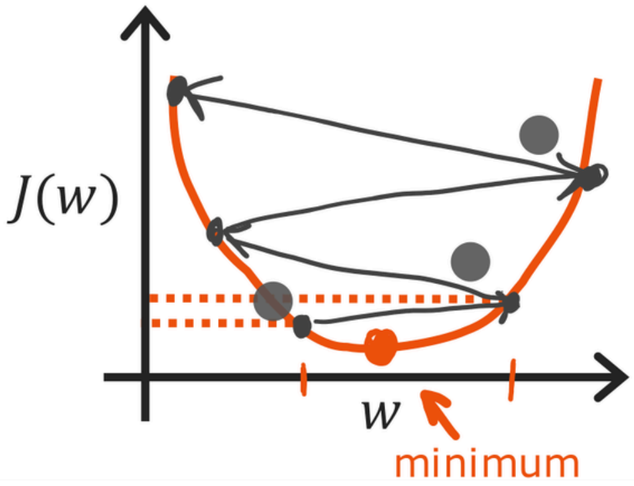
If α is too small...

Gradient descent may be slow.

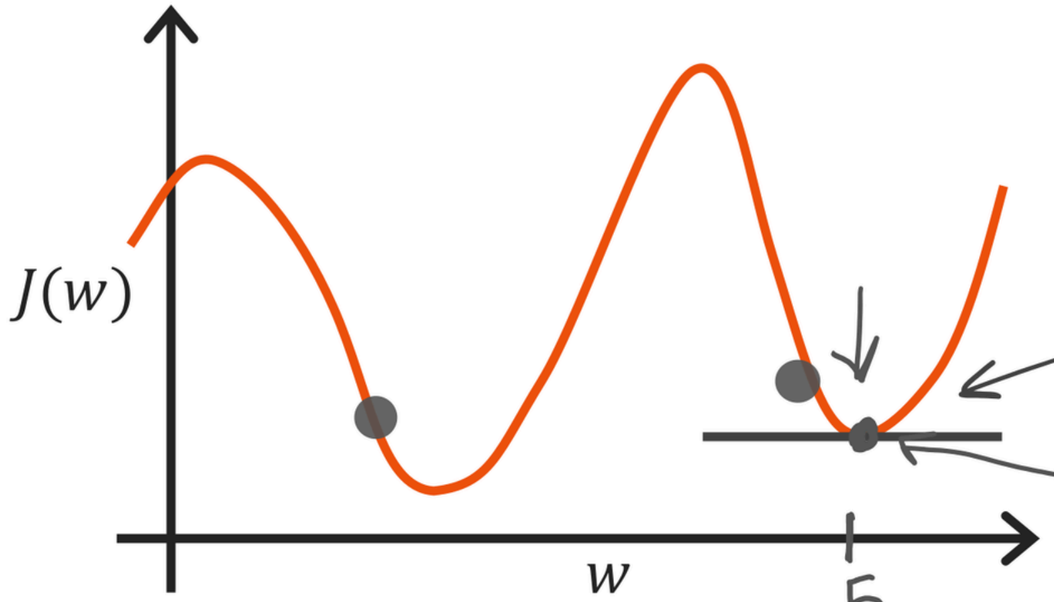
If α is too large...

Gradient descent may:
- Overshoot, never reach minimum
- Fail to converge, diverge





Why Updates Stop at a Minimum



slope = 0

local minimum

$$w = w - \alpha \frac{d}{dw} J(w)$$

0.1

$= 0$

$5 - 0.1 \cdot 0$

$w = w - \alpha \cdot 0$

5 5

$w = w$

Smaller Steps near the Minimum

Can reach local minimum with fixed learning rate α

$$w = w - \alpha \frac{d}{dw} J(w)$$

smaller

not as large

large

Near a local minimum,
- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α

